

Semantic web

The concept of adding intelligent structure to the content of the web as first proposed by Tim Berners-Lee in the 90s. It never took off in the way he proposed.

According to Lotito, his methods can bring greater precision to the accuracy of search results. He claimed that the accuracy of results from purely algorithmic engines is of the order of 30 per cent – in other words, for every meaningful hit, you get three or four results that are just data noise. He claimed that with better automated organisation of the information that the precision could become around 70 per cent, but with some human intervention to validate and organise the data, you can come close to 100 per cent precision. Of course, this will not be possible every time.

But it will be possible when you have closed data sets of information, and if you begin to put many closed data sets with the same kind of organisation, you can get better results. So this is highly applicable to intranets, social networks – any closed set of data.

For this reason, the first uses of this system will not be a publicly accessible general search engine – although that will come – but the engine embedded in some system such as a social network, university web site, government web site, scientific research database, linked network of sites with a common theme, and so on. There are many possibilities.

Lotito agreed with my comment that Facility sounded like an advanced engine for LASEs (limited area search engines), but added that these could be combined together, eventually covering most of the web. I got the strong impression that he wants to set a standard here.

He made the point that “The problem



Gianpiero Lotito and Mariuccia Teroni, the two founders of Facility

for all future search engines is to be able to structure the information that you need to search.” It is impossible to provide advanced services in search without structured information, or, structured access to that information. And there lies the difference in his approach to that usually suggested.

It is important to point out that there is no intention here that web sites should themselves be changed in order for their information to be tagged and structured – the methods of Facility lie in creating a structured approach to information on the web. Part of that structuring process lies in assessing the reliability of sites, and this can only be done by human intervention. This also changes the ranking of sites – engines such as Google rank mainly on the basis of the number of links to various sites, but this is completely blind to issues such as reliability.

Facility employs a pre-filtering engine which reduces the number of potential sites, and then the experts judge those that are the most reliable from the

results. Their starting point is the ODP taxonomy of the internet. The Open Directory Project is an ongoing attempt at cataloguing the contents of the web, and claims to be “the most comprehensive human-edited directory of the Web, compiled by a vast global community of volunteer editors.”

Going down a level of detail, there is also built into the system an ability to catalogue structured sites automatically. This is called the Automatic Taxonomy Extractor, and would be useful for such things as blogs, databases, perhaps news sites, and so on. Sites where there is a regular format to the information. The engine will produce for you an automatic taxonomy, which you can then go in and edit – discarding mistaken headings and tags, adding or combining others, and so forth. This is an example of how the human editing function is aided by software that performs the first part of the work.

This can also work in two ways. If I am developing a search engine for a particular field, then the taxonomy that is developed in this way does not, of course, change the entries in the target site. But, if it is my own site that I am preparing, then I could very well use this tool as a way of tagging directly. This does more than just tag individual items – the overall collection, on say a blog, is structured because the taxonomy is developed and stored. Many people are currently tagging posts on such sites, but without having an overall scheme. This system can propose such a scheme for you, which you can then edit and improve. It will also read those existing tags if they are there.

So, with Facility you have a two-way connection between the engine and the contents of the web. Of course, this is not the semantic web of Tim Berners-Lee that grows from the bottom up. Instead, this is a pragmatic approach, because the tagging of all information on the web in existing web pages is simply not possible. It will never happen.

For Lotito, the display of information in a structured manner is as important as the way in which the results are obtained. The interface is complex at first sight, but easy to use.

The interface is entirely based on Web 2.0 and all the technologies used are open source – the engine can be embedded in other internet systems without licence problems.

The display screen is divided into several sections – Lotito calls them widgets, perhaps not the best term – which include the different formats of information, sources and media that have been identified by a search. These are tiled panels on the screen, in three



Facility structures, organizes access to information being searched and then displays the results as the widgets seen here